

Phishing Email Incident Report

SOC Investigation Summary

Executive Summary

A suspicious email was investigated using header analysis, threat intelligence, and IOC extraction. The message used impersonation and urgency. Header evidence exposed an originating IP and inconsistent sender information. Threat intelligence added reputation evidence. The combined findings support classifying the message as phishing.

Incident Details

Activity	Phishing email investigation
Sender	security@paypal-alert.com
Originating IP	45.33.32.156
Tools	MXToolbox and VirusTotal
Threat Intelligence	4 of 94 vendors flagged the IP as malicious
MITRE ATT&CK	T1566.002 (Spearphishing Link)

Header Findings

Header analysis identified suspicious sender information, an originating IP of 45.33.32.156, and a Return Path that did not match the visible sender information. The mismatch was treated as supporting evidence rather than proof by itself.

Threat Intelligence Findings

VirusTotal showed that 4 of 94 security vendors flagged 45.33.32.156 as malicious. Infrastructure information associated the IP range with Akamai Connected Cloud. Reputation and ownership were used as context and were not treated as proof by themselves.

Indicators

The investigation documented sender information, the originating IP, the Return Path mismatch, suspicious domains and URLs, and urgent account themed language for further searching and response.

Analyst Conclusion

The combined email, header, infrastructure, and threat intelligence evidence supports classifying the message as phishing. The verdict was based on multiple related findings rather than one suspicious characteristic.

Recommended Response

- Mark the message as malicious after validating the evidence.
- Block confirmed malicious domains and URLs.
- Search for other messages containing the same indicators.
- Search for other activity involving the originating IP.
- Determine whether any recipient interacted with the suspicious link.
- Escalate if user interaction or additional suspicious activity is identified.

What I Would Improve

I would extend the investigation by searching for the same indicators across other email and network telemetry. I would also determine whether any recipient interacted with the suspicious link so the investigation could measure actual impact.